PERSONAL RESUME

YANG XINYAN



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Date of Birth: Jan 2006 Location: Shanghai, China

EDUCATION

Sep 2023 – Jun 2027 | **Donghua University (Project 211; University under the Ministry of Education)** — B.S. in Mathematics and Applied Mathematics

- **Academic Performance:** GPA: 3.75/5.0; major rank: 17/68. Admitted to a direct-entry Ph.D. program in Computational Mathematics at Donghua University.
- **Selected Coursework:** Data Science: Technology and Applications (99); MATLAB Mathematical Experiments (96); Data Mining (93); Numerical Analysis (92); Functional Analysis (100); Advanced Algebra (94).

HONORS & TECHNICAL SKILLS

- **Awards:** Donghua University First-Class Scholarship; National Third Prize, 16th China Undergraduate Market Survey and Analysis Competition; National Third Prize, APMCM; National Second Prize, KPMG Business Competition.
- **Certificates:** College English Test Band 6 (CET-6): 514; Putonghua Proficiency Test: Level 2-A.
- **Programming & Analytics:** Python (pandas, scikit-learn), SQL, MATLAB; SPSS for statistical analysis.
- **Methods:** Logistic regression (Logit), K-means clustering, Isolation Forest anomaly detection, and Latent Dirichlet Allocation (LDA) for text mining.
- **Visualization:** Power BI and Excel for multidimensional data visualization and business reporting.

PROJECT EXPERIENCE

Nov 2025 – Apr 2026 | **16th China Undergraduate Market Survey and Analysis Competition** — Core Team Member | National Third Prize

- **Data Preparation:** Led collection of 802 valid survey responses and four in-depth interviews on museum cultural and creative products in Shanghai; conducted variable coding, descriptive statistics, reliability and validity assessments, and randomness tests; scraped and cleaned social media comments using Python.
- **Modeling & Analysis:** Independently built a binary logistic regression model to identify key purchase drivers; developed a dual-path conversion funnel; performed NLP-based text mining in Python; combined K-means clustering with PCA to identify three consumer segments.
- **Deliverables:** Produced a comprehensive research report and visualizations; formulated actionable recommendations for product design and targeted marketing based on differentiated customer profiles.

Oct 2025 – Nov 2025 | **Coffee Shop Annual Revenue Analysis and Forecasting (Kaggle)** — Core Team Member | Course Project: 99/100

- **Exploratory Analysis:** Cleaned a full year of coffee-shop operating data from Kaggle; engineered features, conducted exploratory data analysis (EDA), and developed eight visualizations of sales trends and key variables.
- **Predictive Modeling:** Built SARIMA and Random Forest forecasting models; optimized features and hyperparameters. The tuned Random Forest achieved test-set R^2 of 0.9688, MAE of 21.82, and RMSE of 24.71.
- **Recommendations:** Delivered a sales forecasting report and actionable recommendations for store operations and inventory planning based on model predictions and feature importance.

Nov 2025 | **15th Asia and Pacific Mathematical Contest in Modeling (APMCM)** — Core Team Member | National Third Prize

- **Model Development:** Constructed joint coordinate systems using Denavit–Hartenberg (D–H) parameters and a center-of-mass compensation model for the Unitree G1 humanoid robot; formulated constrained nonlinear motion planning using Lagrangian dynamics, including joint limits and torque constraints.
- **Optimization:** Applied Sequential Quadratic Programming (SQP) for numerical optimization and trajectory planning; compared simulated solutions to evaluate algorithm effectiveness.

Sep 2024 – Dec 2024 | **KPMG Empowerment Program for Women in Engineering (Business Competition)** — Core Team Member | National Second Prize

- **Data Engineering:** Processed 2021–2023 financial data for 76 A-share listed integrated circuit companies from the Wind database using Python; cleaned multiple data sources and engineered 39 features (24 raw variables and 15 derived indicators).
- **Anomaly Detection:** Developed an Isolation Forest model; selected features and tuned parameters using business context; identified four companies with anomalous financial characteristics and analyzed their risk indicators.
- **Reporting:** Created 11 analytical charts in Power BI and a 24-page data analysis report; proposed practical improvements to a digital intelligence platform for audit processes.

RESEARCH EXPERIENCE

Jan 2026 – May 2026 | **Consensus Control of Multi-Agent Systems** — Second Author | Manuscript in Preparation

- **Control Design:** Contributed to an adaptive-scale group bipartite consensus framework for second-order multi-agent systems with cooperative–competitive interactions; designed adaptive parameter update laws based on local consensus errors and developed mathematical derivations and stability proofs.
- **Numerical Simulation:** Independently implemented MATLAB simulations, processed agent-state data, and visualized spiral convergence in phase space and multidimensional error trajectories to validate the proposed control scheme.

Jun 2025 – Jun 2026 | **College Students' Innovation and Entrepreneurship Training Program** — Core Team Member | Shanghai Municipal-Level Project

- **Literature Review:** Independently reviewed more than 50 Chinese- and English-language research papers; created detailed mind maps for 11 studies and synthesized core research threads from 14 recent papers.
- **Mathematical Analysis:** Closely examined six foundational papers, reproduced theorem-level derivations and proofs, and strengthened theoretical understanding of advanced algorithms and rigorous research practices.

CAMPUS & COMMUNITY EXPERIENCE

Jul 2024 – Aug 2024 | **"Return to Hometown" Social Practice Program** — Administrative Assistant | Social Assistance Division, Civil Affairs Bureau

- **Data Verification:** Reviewed over 200 minimum-living-standard assistance applications and participated in home visits to 30 households; checked supporting documentation, indicators, and records for consistency and accuracy.
- **Research & Outreach:** Co-authored two field research reports and suggested process improvements; helped organize eight fire-prevention and drowning-prevention awareness events reaching more than 1,200 residents and students.

STRENGTHS

- **Quantitative Foundation:** Direct-entry Ph.D. admittee with strong foundations in probability, statistics, calculus, and mathematical modeling; able to assess algorithm assumptions and validity alongside hands-on implementation.
- **Learning & Collaboration:** Rapidly distill complex literature and unfamiliar business contexts into tractable analytical questions; apply self-directed learning, problem-solving, and teamwork under tight deadlines.